
Presentation 5

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Goal of this presentation



Main Menu Key 3 vs 8 Redundancy Analysis



Family Menu Chatbot Redundancy Improvement Analysis



Other Legal Menu Design & Queue-reach time analysis

Reminder of Main Menu Key 3 and 8 Redundancy

Observation

Key 3 and option 1 of Key 8 have the **same call flow** (31 sec difference) * LegalMenu1 Prompt is 34 sec

Key 3: If you are calling for help with a legal problem that you have not talked to us about yet, press 3

Key 8: Are you calling for help with a legal problem? If yes press 1, if no press 2

→ Both options asking if the issue is a legal issue and leads to LegalMenu1

Solution

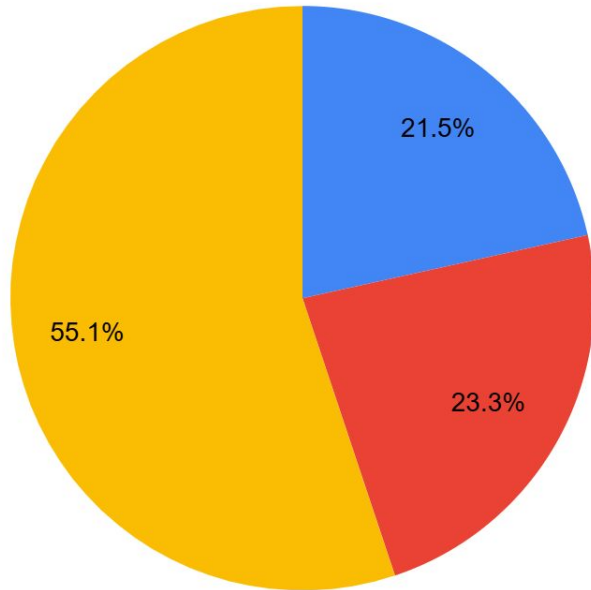
Rephrase Key 3 verbiage to consolidate the two options

Next Steps

Success and path comparison analysis between those that choose key 3 vs key 8

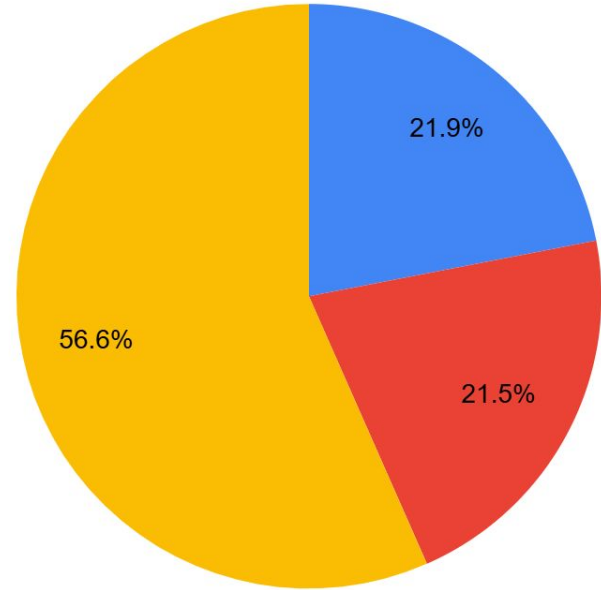
Analysis 1: Outcome Comparison

Pressed Key 3



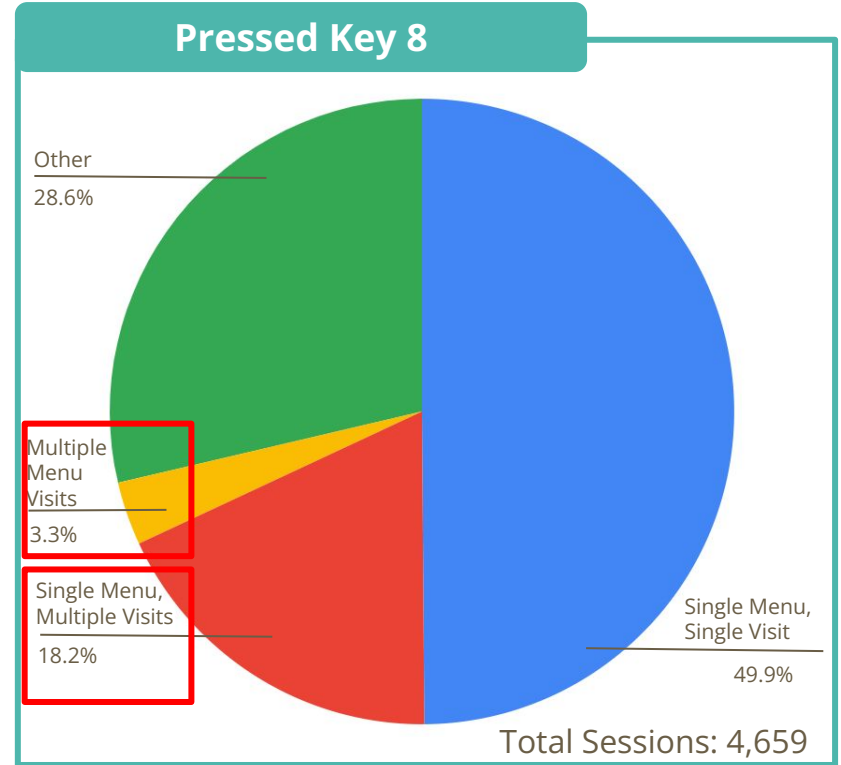
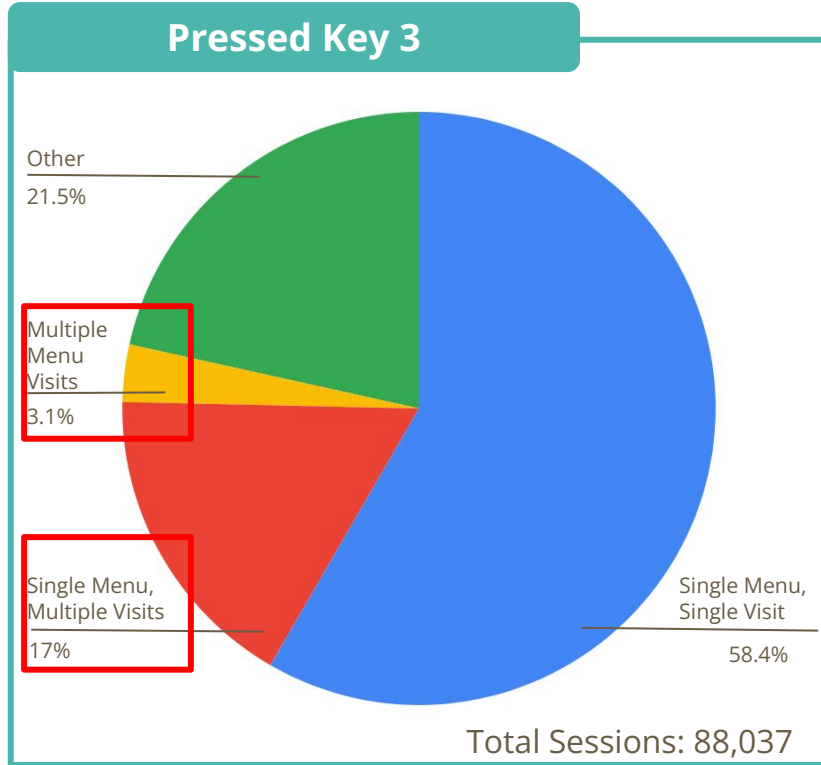
● Positive ● Neutral ● Negative

Pressed Key 8

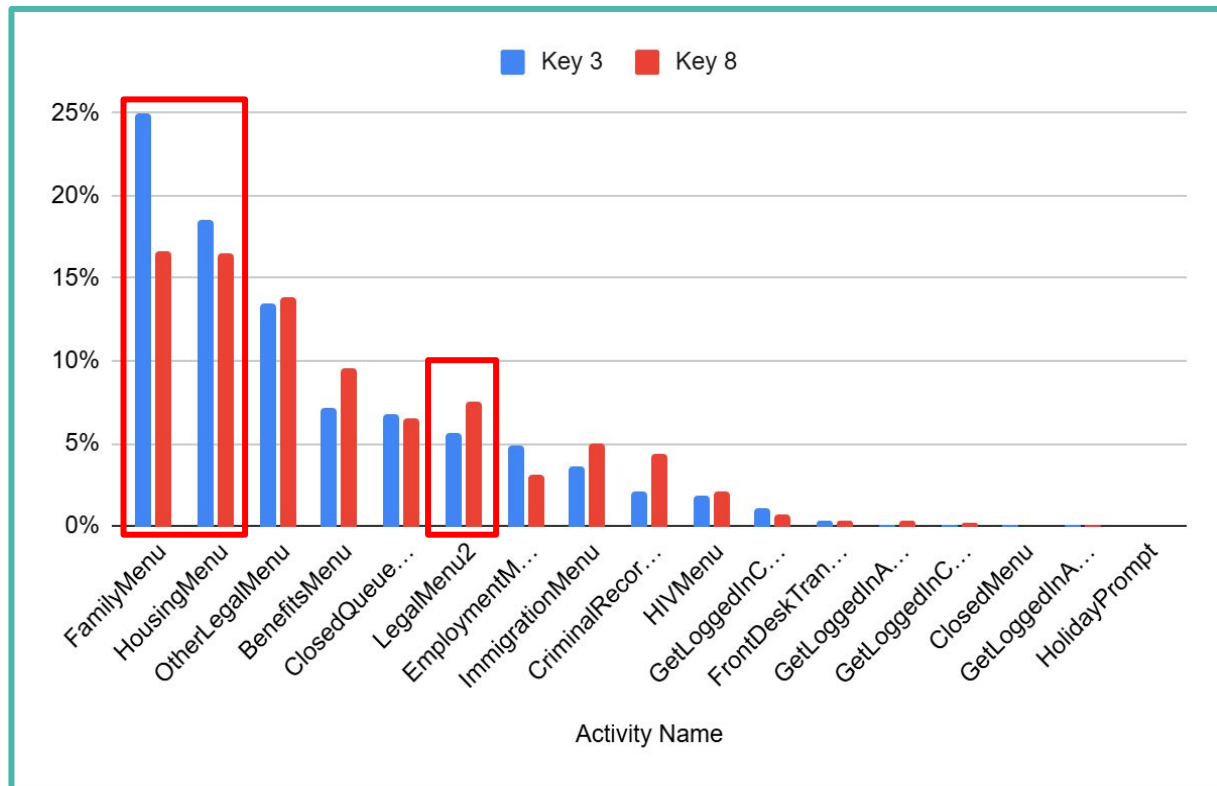


● Positive ● Neutral ● Negative

Analysis 2: Redundancy at Legal Menu Comparison



Analysis 3: First Activity Distribution Comparison



Summary

- 1) Overall distribution is similar across both keys
- 2) Higher proportion of callers repeats LegalMenu2 for those that pressed key 8
→ Those that choose key 8 may be more confused
- 3) Higher proportion of callers choose Family or Housing for those that pressed key 3
→ Key 3 captures the more popular options

Analysis Summary + Recommendation

Analysis Summary

Analysis 1: Similar outcome proportions

Analysis 2: Similar proportion of path repetitions (~20%)

Analysis 3: Key 3 has less LegalMenu2 repetitions and captures a higher proportion of popular options

Solution

Rephrase Key 3 verbiage to consolidate the two options

→ Saves time and simplifies options

FamilyMenu Chatbot Analysis

Question: Did TransferToSafeHaven (chatbot introduced in FamilyMenu from Aug 11, 2025 to Aug 29, 2025) lead to less confusion (redundancies or unsuccessful calls) in the legal menu?

Further, could a chatbot help with other submenus?

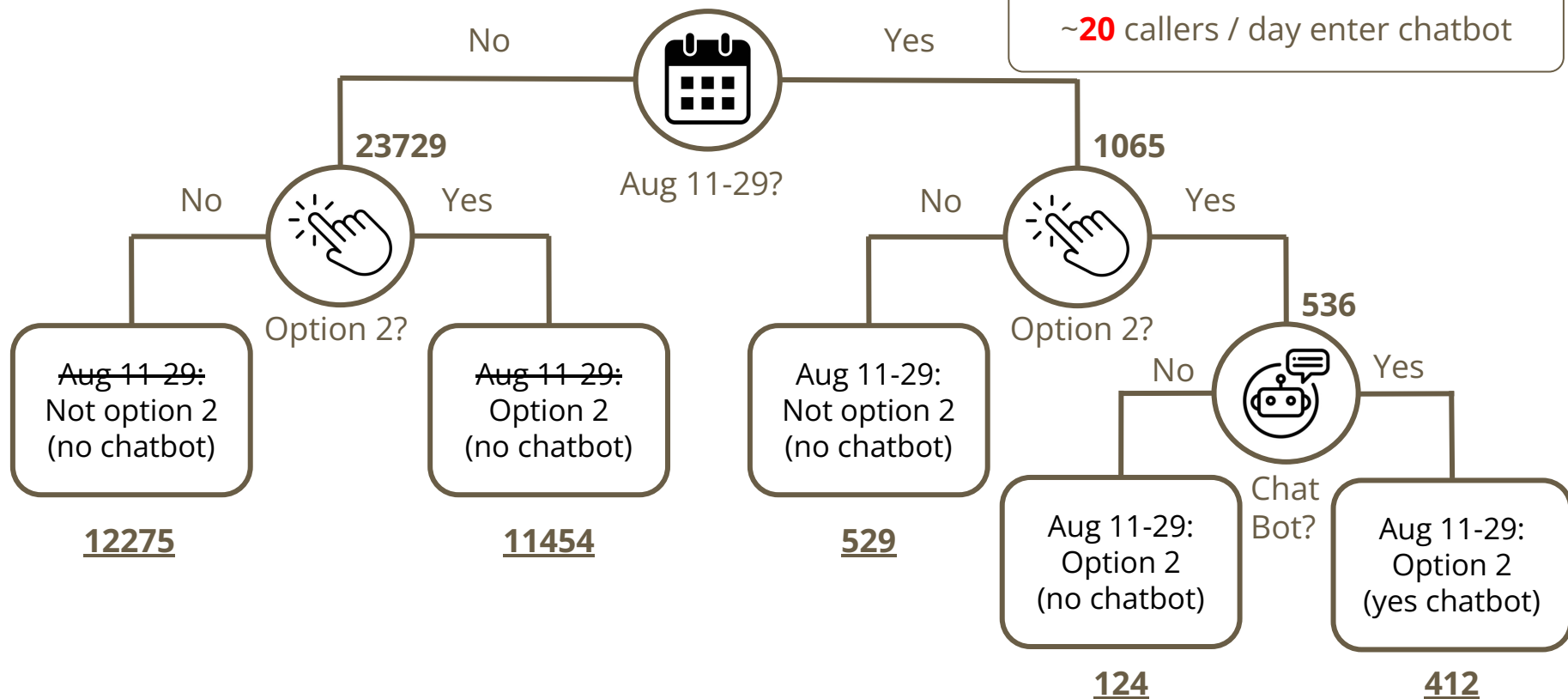
- Before/After Aug 11-29: **Option 2 -> DivorceOrParentingMenu**
- Aug 11-29: **Option 2 -> Directly to Chatbot**
 - Other options unchanged
 - 1,3,5 -> FamilyQueue
 - 4 -> EducationQueue
 - 9/Invalid/No -> LegalMenu

FamilyMenu Visit Counts

~**50** callers / day visit FamilyMenu

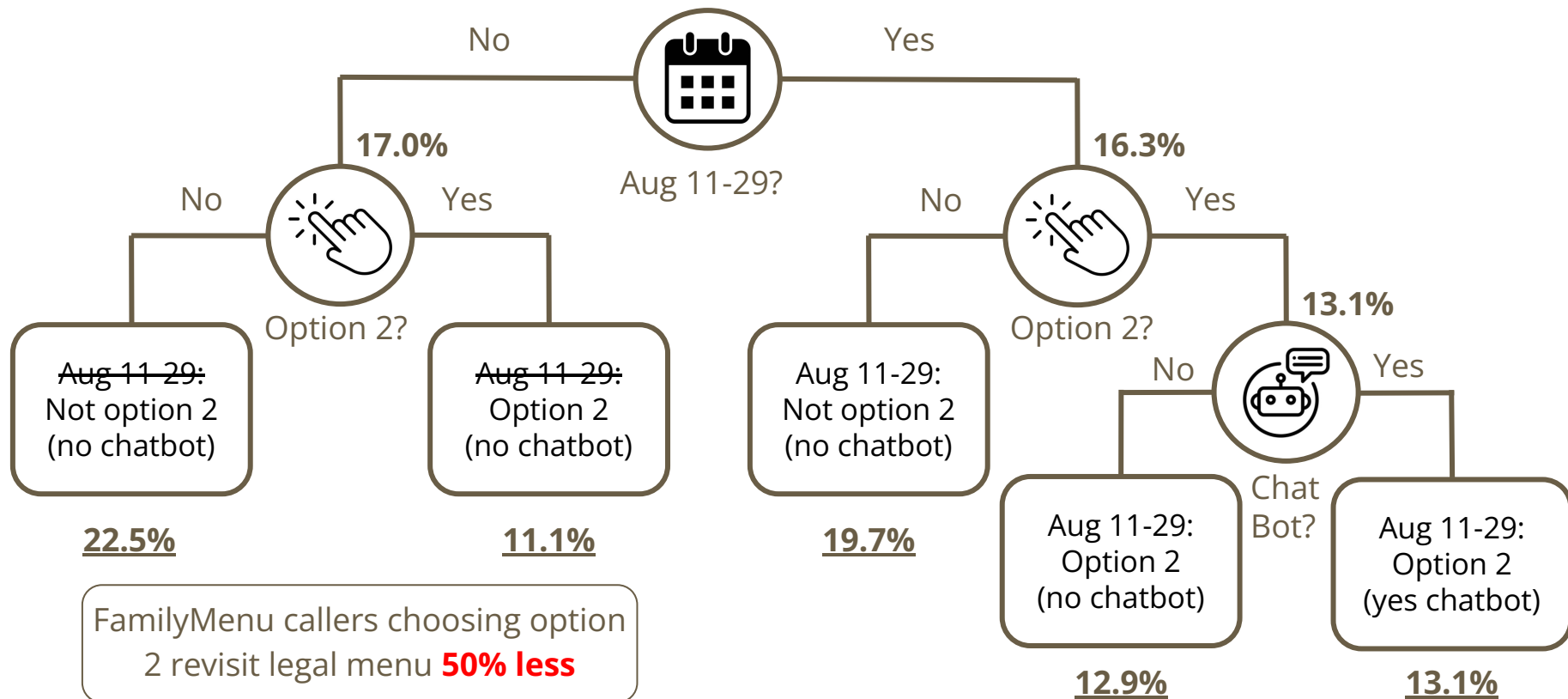
~**50%** press option 2 at some point

~**20** callers / day enter chatbot



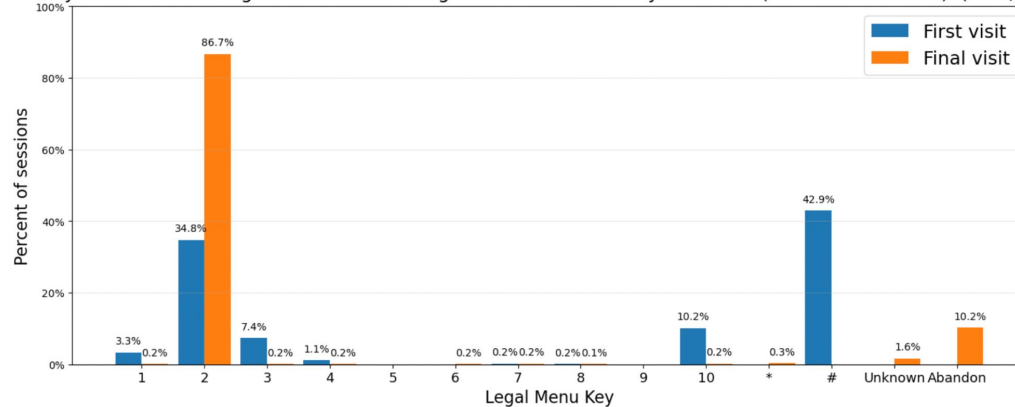
FamilyMenu Redundancies (% with >1 visits to Legal Menu)

Chatbot **did not impact** legal menu
revisit rate for those choosing option 2

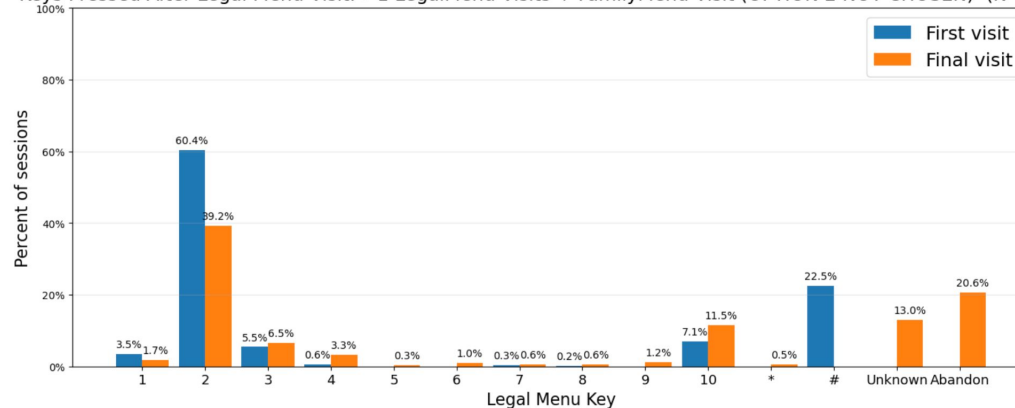


Redundancy Analysis (FamilyMenu Callers)

Keys Pressed After Legal Menu Visit: >1 LegalMenu Visits + FamilyMenu Visit (OPTION 2 CHOSEN) (N=1,289)



Keys Pressed After Legal Menu Visit: >1 LegalMenu Visits + FamilyMenu Visit (OPTION 2 NOT CHOSEN) (N=2,860)



1. HIV
2. Family
3. Housing
4. Benefits
5. Consumer(SP) Queue
6. Criminal Records Voicemail
7. Employment
8. Immigration
9. ADAPT(SP) Queue
10. Other Legal

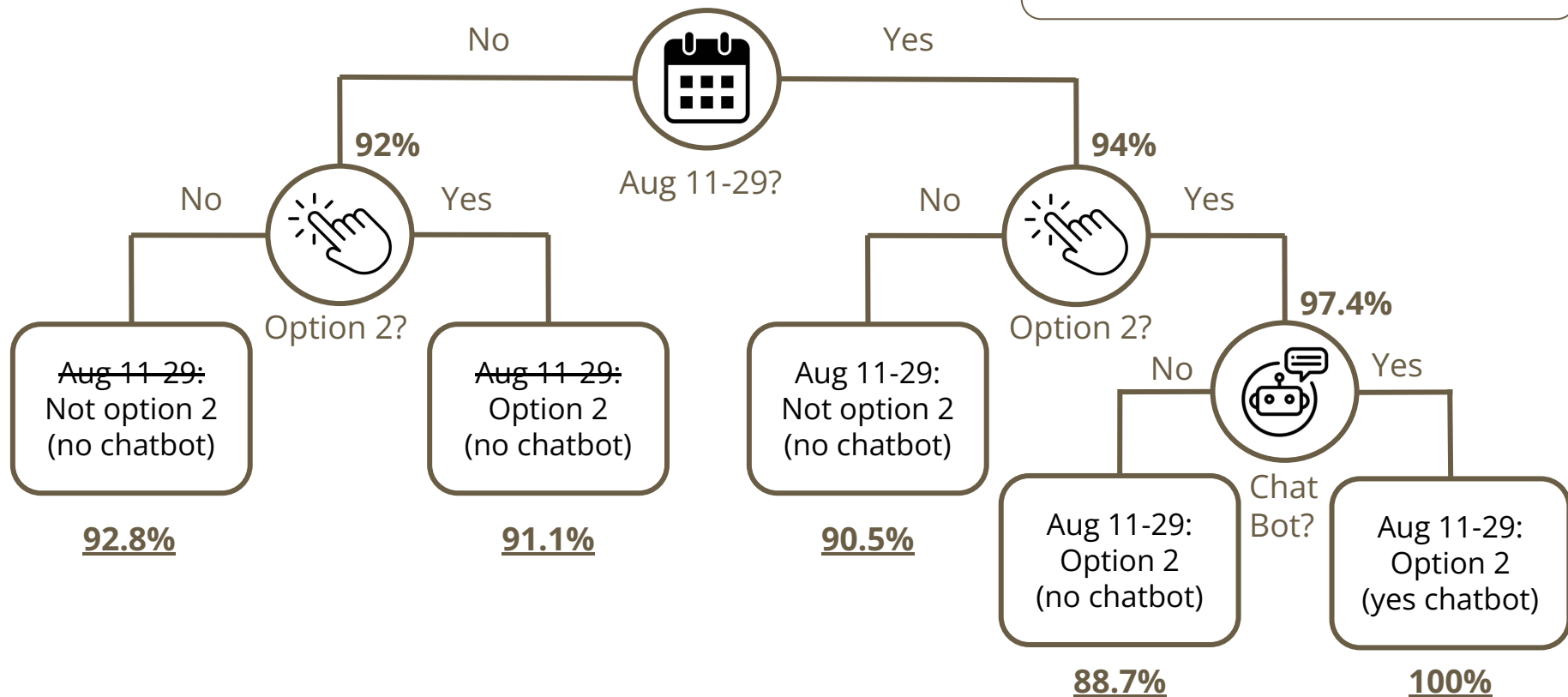
#/no/invalid input.
Legal Menu (again)

*. Front Desk Transfer

Unknown. Closed Queue

FamilyMenu % with 'Successful' Calls

Chatbot **marginally improves** call success rate



Summary / Steps for Next Quarter

- FamilyMenu Chatbot has little impact on reducing LegalMenu confusion
 - Redundancies / confusion come before visiting DivorceOrParenting submenu
 - For a chatbot to help reduce confusion, this would need to come predominantly after visiting submenu (ex: right after LegalMenu -> FamilyMenu)
 - Slightly improves 'success' rate for calls
 - Especially better for submenus with lower success rates (like Housing submenu)
- Alternative Chatbot Usage: OtherLegalMenu
 - Collect data on what issues fall under OtherLegalOtherMenu

Other Legal Menu Recall

Issue

Issue 1: The verbiage states “We can’t help you” in all 3 submenus

Issue 2: Significant numbers of callers are not hearing the full prompt
→ may be missing important information that may help them.

Issue 3: Other Legal Sub-menus are dense with information. Even if they hear the full prompt, how many would actually remember this information?

Solution

Solution 1: Acknowledge limits upfront

Clearly state Legal Aid’s limitation and the availability of other resources at the beginning
→ avoids repeating it each time in each sub menu.

Solution 2: Issue-specific options

Let callers select the type of legal issue they have and provide tailored resources (website, hotline, clinic info)
→ helps caller prepare for information that will be thrown at them.

Solution 3: Optional SMS delivery

Offer a text message option so users can receive resources directly on their phone

- From a design perspective, you should never expect a user to have to remember a significant amount of information

**In the next couple of slides,
I propose a new design for Other Legal**

Old Verbiage

IVR-OtherLegalMenu-English.wav	2023	Other Legal Menu	If you are calling about a criminal case, press 1. If you are calling about a personal injury, car accident, traffic court, or medical malpractice, press 2. For all other issues, press 3. (goes to Menu 2027). To repeat these options, press the pound key. To return to the previous menu, press 9.
IVR-OtherLegalCriminalCase-English.wav	2025	Criminal	We do not handle criminal cases. If you have internet access, you can visit illinoislegalaid.org for information and referrals. You can also call CARPLS, a free legal advice hotline, at 312-738-9200. To repeat this message, press the pound key.
IVR-OtherLegalPersonalInjury-English.wav	2026	Personal Injury	We do not handle personal injury, car accident, traffic court or medical malpractice cases. If you have internet access, you can visit illinoislegalaid.org for information and referrals. You can also call CARPLS, a free legal advice hotline, at 312-738-9200. To repeat this message, press the pound key.
IVR-OtherLegalOther-English.wav	2027	Other Legal Other	Unfortunately we have limited resources and can only provide help with certain types of cases. If you have internet access, you can visit illinoislegalaid.org for information and referrals on many common legal issues. You can also call CARPLS, a free legal advice hotline, at 312-738-9200. Finally, you may be able to meet with a free lawyer to get advice about your legal issue at one of our monthly neighborhood advice clinics. For information about upcoming clinic dates and locations, press 1. To repeat this message, press the pound key. To return to the previous menu, press 9.

- Other Legal Menu is the parent menu of the other three.
- We will propose a new menu structure

Other Legal Menu

Old: If you are calling about a criminal case, press 1. If you are calling about a personal injury, car accident, traffic court, or medical malpractice, press 2. For all other issues, press 3. (goes to Menu 2027). To repeat these options, press the pound key. To return to the previous menu, press 9.

New: While we do not handle Other Legal Cases, we can provide additional resources. For additional resources, press 1. To repeat, press the pound key. To return to the previous menu, press 9.

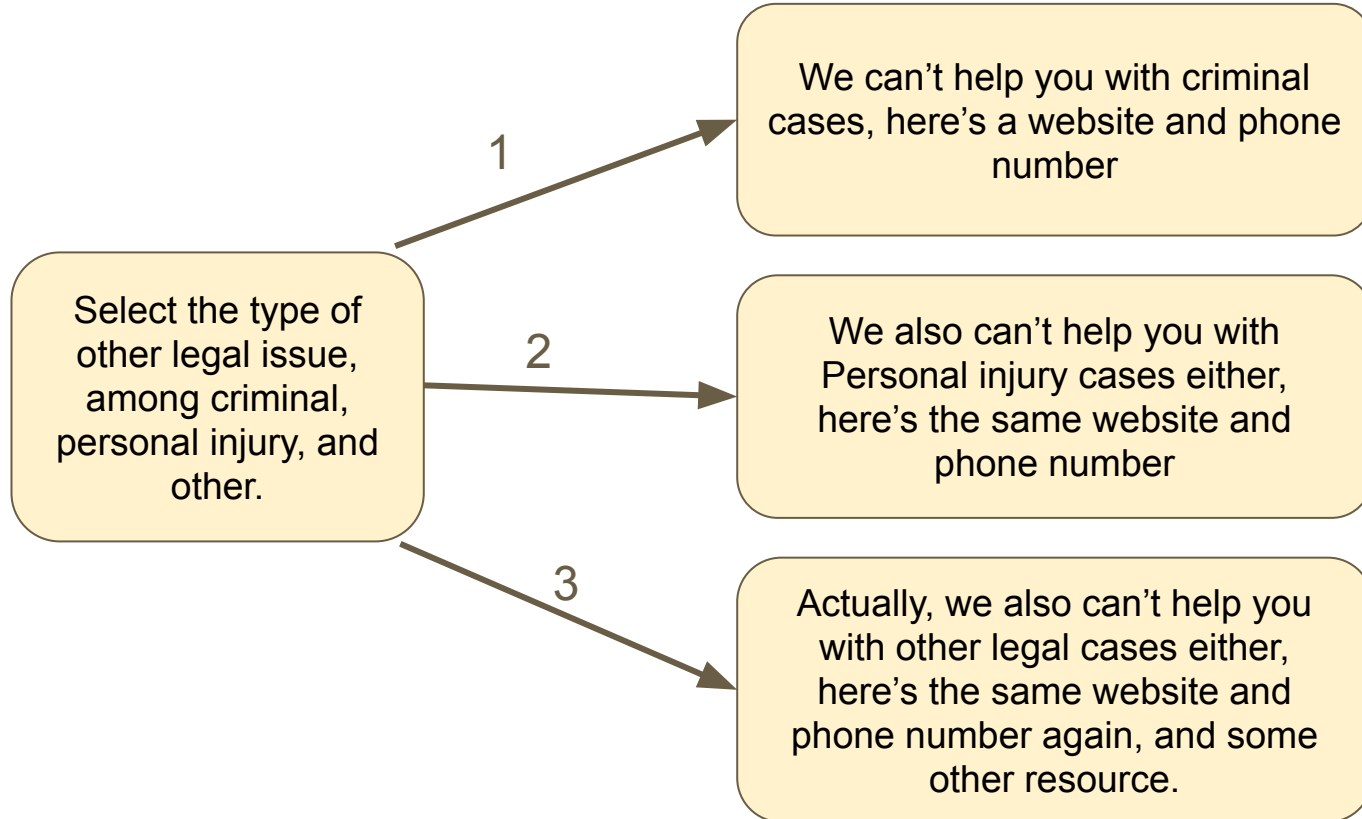
Other Legal Submenus:

Old:

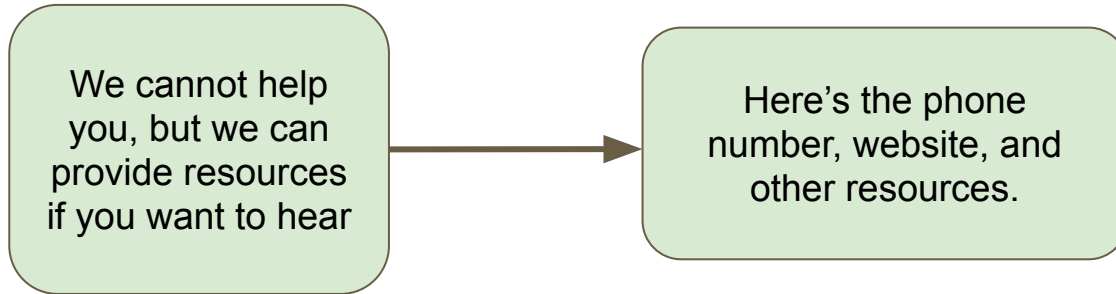
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New (becomes only one menu): If you have internet access, you can visit illinoislegalaid.org for information and referrals. You can also call CARPLS, a free legal aid hotline, at 312-738-9200. Finally, you may be able to meet with a free lawyer to get advice about your legal issue at one of our monthly neighborhood advice clinics. For information about upcoming clinic dates and locations, press 1. To repeat this message, press the pound key. To return to the previous menu, press 9.

Old Flow Chart



New Flow Chart



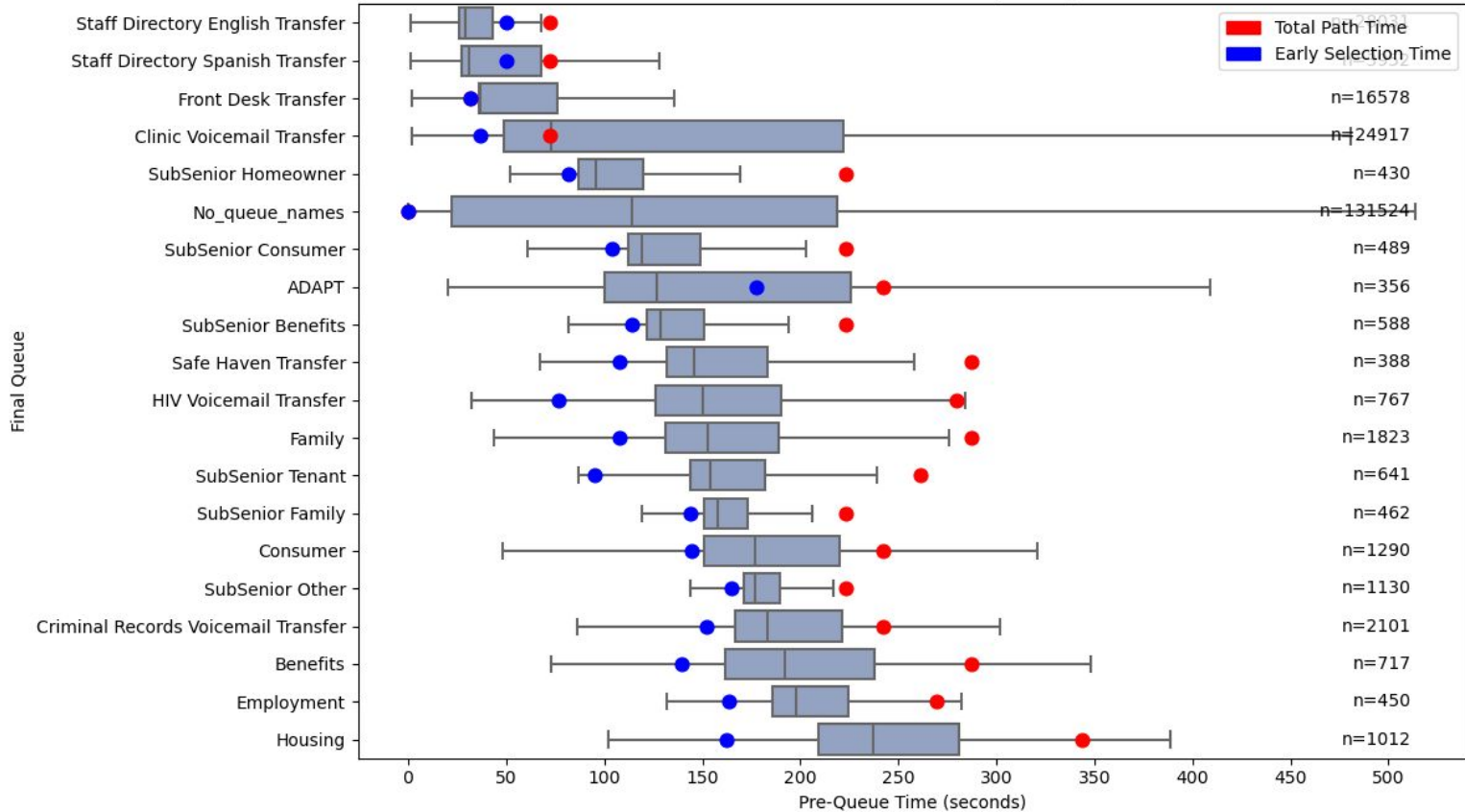
Time Distributions Analysis

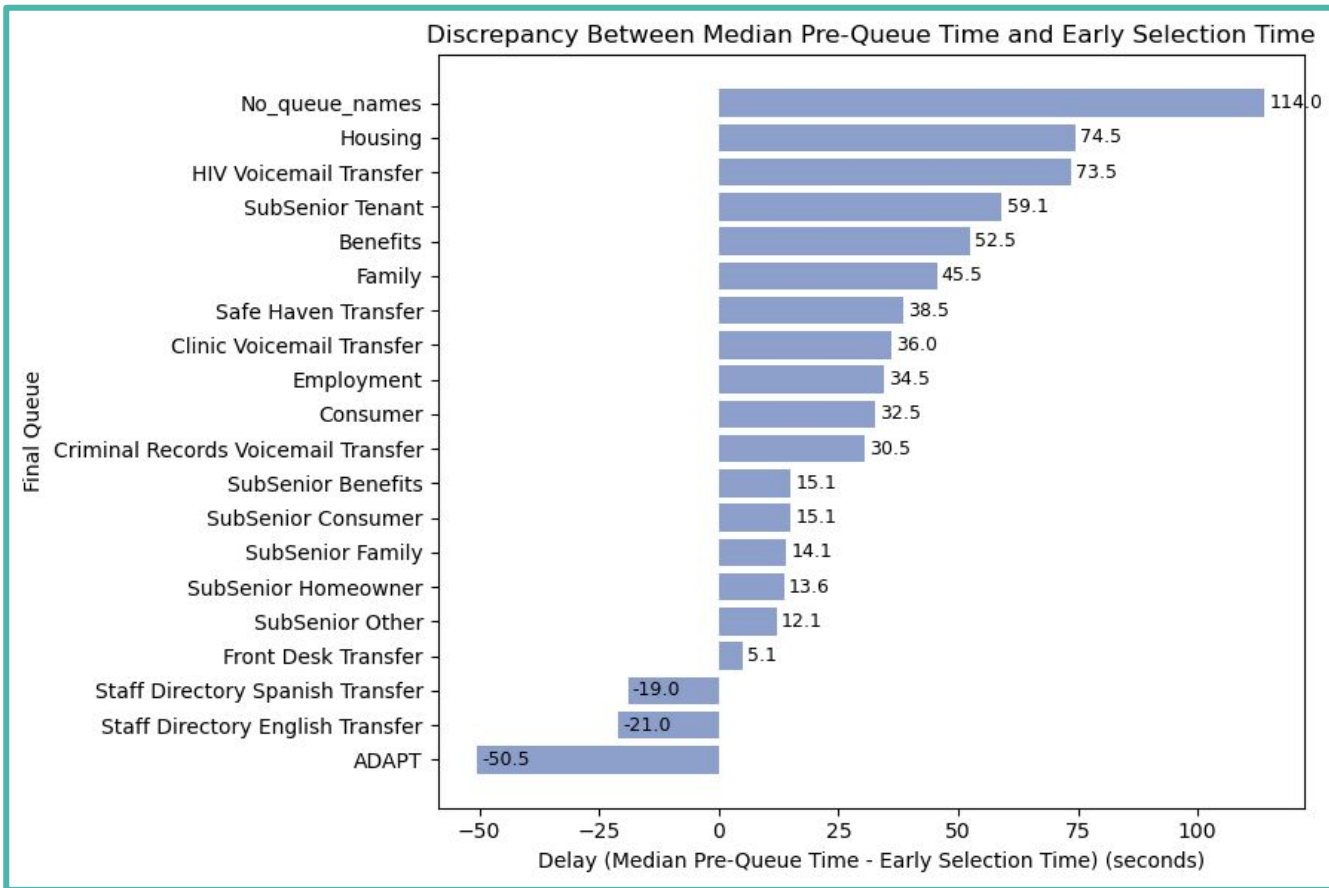
- Last presentation, I made a barplot on the time it takes to reach each final queue for the top 20 most common queues.
- For simplicity, I will only consider calls starting from the Main Number EP
- I added two additional data points for each final queue
 - Total Path Time
 - Early selection Time
- These times are meant to be an estimate of how much time a caller should spend to reach the final queue; We define them in more detail next slide
- If we observe that either of these times is much lower than the median time it took someone to reach that queue in the data, this indicates redundancy and confusion in reaching that queue

Background and definitions

- **Total Path Time:** total time it would take for the shortest path (fewest nodes) to reach the queue, assuming prompt is fully heard
 - Calculated by summing the durations of all prompts along the shortest path
 - For prompts without a recording, the duration is estimated from the number of words
- **Early selection Time:** total time it would take for the shortest path to reach the queue, assuming users select the correct option to the queue right after they finish hearing it.
 - Calculated similarly, but each prompt's time only includes the portion until the user sees the option leading to the correct final queue
- There may be better ways to calculate the “ideal” path time
 - Given my limited time for this project, I only covered these two.

Distribution of Pre-Queue Time with Corresponding Final Queue Times





Future steps

- The previous slides show us the queues with the longest delays (**Housing, HIV, and Benefits**)
- These queues are ones where callers are spending much more time than they need in order to reach the queue
- I recommend that future project groups work on:
 - For each queue with high delay, try to figure out the causes
 - Proposing a new way to measure “ideal” path time, if that might be better

Thank You!
